Exercise Sheet 11

Throughout, K is a compact Lie group with normalised Haar measure. The set $\widehat{K}$ is a fixed set of representatives of the equivalence classes of irreducible unitary representations. The Fourier transform and the convolution are

$$\hat{f}(\Pi) = \int_K f(g)\Pi(g^{-1})\,dg, \qquad (f * h)(x) = \int_K f(y)h(y^{-1}x)\,dy.$$

We use the matrix coefficients $c_A(g) = \operatorname{tr}(A\Pi(g))$, the Hilbert–Schmidt inner product $\langle A, B\rangle_{\mathrm{HS}} = \operatorname{tr}(A^*B)$, and the results of Lecture 11. On $\mathrm{SU}(2)$ we write

$$U = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1,$$

and E_{ij} is the matrix unit with 1 in row i and column j. Exercises 1–6 are core. Exercise 7 is an optional extension.

Exercise 11.1 (Matrix units and the order of multiplication). Let $\Pi \in \widehat{K}$ have dimension d and orthonormal basis $e_1, \dots, e_d$. Put $D_{ij}(g) = \langle e_i, \Pi(g)e_j\rangle$.

(a) Show that $D_{ij} = c_{E_{ji}}$, where $E_{ji} = e_j e_i^*$. Deduce $\hat{D}_{ij}(\Pi) = d^{-1}E_{ji}$.

(b) Use the convolution theorem to prove $D_{ij} * D_{kl} = d^{-1}\delta_{jk}D_{il}$.

(c) Explain why the functions multiply in the natural order while their transforms multiply in reverse.

Exercise 11.2 (A direct $\mathrm{SU}(2)$ computation). Let Π be the defining representation of $\mathrm{SU}(2)$ on $\mathbb{C}^2$.

(a) Show that $\int |a|^2 = \int |b|^2 = \frac{1}{2}$ and $\int a\bar{b} = 0$. Using right translation by $t = \operatorname{diag}(i, -i)$, show that $\int ab = \int b^2 = \int \bar{a}\bar{b} = \int \bar{b}^2 = 0$.

(b) Let x have entries α, β in the pattern of U. Compute $(U_{11} * U_{12})(x)$ and $(U_{12} * U_{11})(x)$ directly from the definition.

(c) Compare with Exercise 11.1. With $f = U_{11}$ and $h = U_{12}$, show that the wrong order $\hat{f}\hat{h}$ would contradict (b).

Exercise 11.3 (Plancherel for a coefficient of the adjoint representation). Let $V = \mathfrak{sl}(2, \mathbb{C})$ with $\langle X, Y\rangle = \frac{1}{2}\operatorname{tr}(X^*Y)$, and let $\mathrm{Ad}(U)X = UXU^{-1}$ for $U \in \mathrm{SU}(2)$. Let $\sigma_3 = \operatorname{diag}(1, -1)$ and $f(U) = |a|^2 - |b|^2$.

(a) Show that Ad is unitary and irreducible, hence equivalent to V_2. You may use that the adjoint representation of $\mathfrak{sl}(2, \mathbb{C})$ is irreducible.

(b) Show that $f(U) = \langle \sigma_3, \mathrm{Ad}(U)\sigma_3\rangle$.

(c) Compute $\hat{f}$ at Ad, and compute $\|f\|^2$ by Plancherel.

(d) Check (c) independently. You may use the following facts. The map $U \mapsto (a, b)$ carries Haar measure to normalised surface measure on $S^3 \subset \mathbb{C}^2 = \mathbb{R}^4$, with coordinates $a = x_1 + ix_2$ and $b = x_3 + ix_4$. For this measure, $\int x_1^4 = \frac{1}{8}$ and $\int x_1^2 x_2^2 = \frac{1}{24}$.

Exercise 11.4 (A central kernel on $\mathrm{SU}(2)$). Let $k(U) = |\operatorname{tr} U|^2$ and $T_k f = f * k$ on $C(\mathrm{SU}(2))$.

(a) Show that k is central and $k = \chi_0 + \chi_2$.
(b) Show that $T_k = P_0 + \frac{1}{3} P_2$.
(c) Compute $T_k f$ for $f(U) = |a|^2$.
(d) Check the value of $T_k f$ at $U = I$ directly, with the moments of Exercise 11.3(d).

Exercise 11.5 (Comparison with the circle). Let $K = \mathrm{U}(1)$ with $dg = d\theta/2\pi$ and $\chi_n(e^{i\theta}) = e^{in\theta}$. As in the lecture, write $\hat{f}(n)$ for $\hat{f}(\chi_n)$ and P_n for P_{χ_n}.

(a) Show that $f * h = h * f$ for $f, h \in C(\mathrm{U}(1))$, that every convolution operator is central, and that $P_n f = \hat{f}(n)\chi_n$.
(b) Write $f(\theta)$ for $f(e^{i\theta})$. Let f be twice continuously differentiable as a function of θ. Show that $|\hat{f}(n)| \le \sup |f''|/n^2$ for $n \neq 0$. Deduce that its Fourier series converges absolutely and uniformly to f.
(c) Does (b) extend to all continuous f?

Exercise 11.6 (Projector identities). Let $\Pi, \Pi' \in \widehat{K}$.

(a) Prove that $\chi_\Pi * \chi_{\Pi'} = \delta_{\Pi\Pi'} d_\Pi^{-1} \chi_\Pi$, and that $\chi_\Pi * f = f * \chi_\Pi$ for all $f \in C(K)$.
(b) On $\mathrm{SU}(2)$, check (a) for χ_1 at $U = I$ and at $U = -I$.
(c) Show that P_Π commutes with every L_g and every R_g on $C(K)$.

Exercise 11.7 (Optional extension: central functions and sine series). Let $t_\theta = \operatorname{diag}(e^{i\theta}, e^{-i\theta})$. Use the Weyl integration formula of Lecture 9 for central f on $\mathrm{SU}(2)$:

$$\int_{\mathrm{SU}(2)} f \, dU = \frac{2}{\pi} \int_0^\pi f(t_\theta) \sin^2\theta \, d\theta.$$

(a) For central continuous f put $F(\theta) = f(t_\theta)\sin\theta$. Show that $\langle \chi_m, f \rangle = \frac{2}{\pi} \int_0^\pi F(\theta) \sin((m+1)\theta)\, d\theta$.
(b) Show that Plancherel for central f is Parseval's identity for F in the orthonormal basis $\sqrt{2/\pi}\,\sin(n\theta)$, $n \ge 1$, of $L^2([0,\pi], d\theta)$.
(c) Check (a) for $f = |\operatorname{tr} U|^2$ and $m = 0$.